Revisiting Motion-Aware Checkerboard Calibration for Event Cameras

Taehun Ryu, Changwoo Kang, and Kyungdon Joo

***Abstract*—The conventional checkerboard-based calibration for standard cameras faces fundamental limitations when applied to bio-inspired event cameras. These limitations arise from three challenges: (i) Events are triggered asynchronously at different timestamps along motion trajectories. If we accumulate them directly on the image plane, it causes temporal misalignment and produces blurred edges. (ii) Checkerboard corners on event cameras show near-zero event occurrence at the corner itself. (iii) Motion parallel to a checkerboard edge suppresses its event response, limiting the observable corner structure. This hinders reliable corner localization and makes calibration difficult. To address these issues, we present a novel calibration framework that directly detects checkerboard corners from event data without learning-based grayscale image reconstruction. We first mathematically analyze event generation at checkerboard corners and edges to guide corner localization. We then distinguish motion cases using gradient orientation statistics and initialize corners from the corresponding edge cues. Under parallel motion, we recover the suppressed edge using polarity information. Using the near-zero event occurrence at checkerboard corners, we gradually refine the estimated corner toward low event-density regions, achieving sub-pixel accuracy. Furthermore, we extend our framework to AprilTag detection under partial visibility and stereo event camera calibration. Experiments on synthetic and real-world data from event cameras with different resolutions demonstrate reliable corner detection and accurate calibration.**

***Index Terms*—Event camera, camera calibration, checkerboard detection, AprilTag detection**

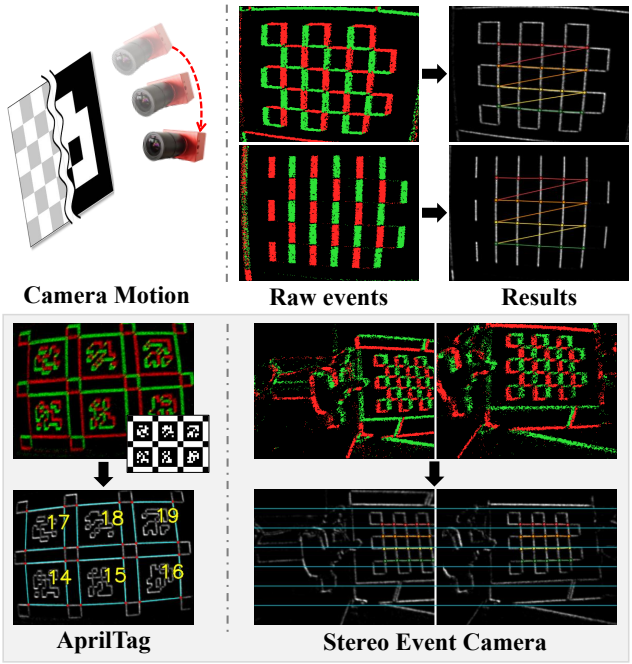


Fig. 1. Illustration of the proposed framework. Our method detects checkerboard corners regardless of motion direction and further extends to AprilTag [15] detection and stereo event camera calibration.

I. Introduction

THE event camera, a neuromorphic vision sensor, has attracted widespread attention for its robustness to motion blur and low-light conditions compared to conventional cameras [1], [2]. Consequently, event-based vision has shown significant effectiveness in diverse computer vision and robotics domains such as 3D scanning [3], robotic manufacturing [4], high-speed object detection and tracking [5]–[8], and autonomous navigation [9], [10].

As the demand for such applications increases, calibration of event cameras becomes crucial for reliable vision systems as a process of estimating camera parameters. For frame-based cameras, the camera parameters are typically determined by minimizing the reprojection error between 2D fiducial points detected from images and their corresponding 3D points pre-defined on the calibration target. Various pre-defined patterns (*e.g.,* checkerboard [11], circular grid [12]) have been developed to provide easily observable fiducial points for accurate calibration. Among them, the checkerboard pattern is the most common and reliable calibration target [13], [14]. Specifically, the fiducial points of the checkerboard are the corners of black-and-white squares, which are defined by the intersections of two edges. These characteristics remain geometrically well-defined under perspective projection or lens distortion.

However, the checkerboard pattern remains unexplored for event cameras, as it cannot be directly applied to a raw event stream. This is because event cameras do not preserve the image structure. Instead, they record relative brightness changes asynchronously at each pixel as binary values. For this reason, some methods utilize learning-based reconstruction to convert a raw event stream into grayscale images, and then apply frame-based calibration to the reconstructed grayscale images [16], [17]. Nevertheless, artifacts of reconstructed images inherently constrain calibration precision. Some methods utilize blinking LED boards to directly process a raw event stream [18], [19]. They generally yield accurate calibration results, but they demand high hardware requirements and rely on dedicated procedures. Other approaches employ circle-grid

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Project page: https://vision3d-lab.github.io/corner2tag++.

patterns for event-based calibration [20]–[22]. They exhibit motion-invariant characteristics and achieve robust detection performance, yet their design inherits the drawback of circular patterns, which are sensitive to lens distortion [12]. Compared to the above approaches, the checkerboard pattern provides geometrically consistent and precisely localized corners, ensuring high calibration performance. It also holds strong potential for extension to ID-based markers such as ARTag [23], and AprilTag [15], [24], as they share a similar checkerboard-based corner detection principle.

In this work, we advocate the revisit of the standard checkerboard pattern using event cameras to preserve its geometric advantages for accurate calibration, while extending the framework to fiducial markers and stereo event camera calibration (see Fig. 1). However, checkerboard detection from event data faces three key challenges. First, due to the asynchronous characteristic of each event, stacking them into a single image produces blurred edges and obscures intersections [25]. Second, events are not triggered at checkerboard corners due to their locally symmetric intensity pattern, which limits reliable corner detection. Third, if checkerboard edges are parallel to camera motion, no events are triggered along that edge [22]. Consequently, only one of the two intersecting edges may remain directly observable, making it difficult to determine the corner from the remaining edge.

To resolve these, we propose the first event camera calibration framework that directly detects checkerboard corners from event representation without grayscale image reconstruction (see Fig. 2). We create an Image of Warped Events (IWE) to temporally align asynchronous events. This process produces a sharp, static representation, which is essential for precise corner detection. Then, we provide a mathematical analysis of Event Rate (ER) at checkerboard corners. This analysis confirms that events are theoretically absent at the exact corner intersections. Grounded in this analysis, our method first initializes corners using strong edge cues, instead of finding corners directly. We then refine the initial corners by moving them toward the point of minimum event density to achieve sub-pixel accuracy. These accurate corner estimates are then utilized to optimize the camera parameters. Furthermore, we demonstrate that our framework is applicable to fiducial markers, such as AprilTag [15], [24]. This enables reliable tag identification even under partial visibility. We also apply the framework to stereo event camera calibration using synchronized event cameras. This yields accurate stereo extrinsics, enabling consistent rectification of the two views.

This paper extends our previous work published in [26]. We generalize the framework by reducing its restriction on motion direction and adding support for stereo calibration. In particular, we further address parallel motion case, where event generation along one checkerbaord edge is suppressed. To handle parallel motion, we introduce a motion distinction module (Sec. VI) and a dedicated corner initialization strategy (Sec. VII-C), enabling reliable corner detection under parallel motion. Moreover, we extend the framework to stereo event camera calibration by temporally aligning the left and right event streams to a common reference time (Sec. IX). We expand our evaluation to event cameras with different resolutions to assess applicability across different sensor configurations. In summary, the contributions of this work are as follows:

- **Event-based checkerboard calibration**: We present the first calibration framework using detected checkerboard corners without grayscale image reconstruction.
- **Mathematical analysis of corner events**: We analyze the absence of events at corner intersections, which we then exploit for corner refinement.
- **Motion-aware corner localization**: We distinguish motion cases using gradient orientation analysis and apply dedicated corner initialization.
- **AprilTag detection and stereo calibration**: We extend our framework to AprilTag detection and stereo event camera calibration using synchronized event cameras.

II. Related Work

Camera calibration process can be divided into two steps: Pattern detection and camera parameter optimization [27]. Pattern detection aims to identify accurate fiducial points that serve as 2D-3D correspondences from predefined calibration targets. Camera parameter optimization then estimates the intrinsic and extrinsic parameters of cameras by solving an optimization problem based on these correspondences. In this section, we mainly deal with pattern detection for both frame-based and event cameras. Please refer to other frameworks [13], [14], [28] for details of the optimization process.

A. Pattern Detection for Frame-Based Cameras

As fiducial points directly provide 2D-3D correspondences, various calibration targets have been proposed [11]–[13], [15]. Among them, the checkerboard pattern [11], [13], [14], [29] serves as the standard for high-precision calibration due to its geometrically well-defined and densely distributed corners, which enable robust and sub-pixel localization even under perspective transformation or lens distortion. Building upon this concept, ARTag [23], AprilTag [15], [24], and ArUco marker [30] encode unique marker IDs to ensure reliable detection under partial observations. Accordingly, the checkerboard remains the most widely adopted target in frameworks such as OpenCV [13]. In parallel, other primitives such as circular features or dots are used for calibration targets [12], [31], [32]. They use centroids of circular shapes as fiducial points. They exhibit rotational invariance but suffer from degraded localization accuracy under nonlinear transformations such as lens distortion [12], [33].

In this work, our objective is to apply the well-validated checkerboard pattern to event cameras while leveraging its proven calibration accuracy and structural extensibility to ID-coded fiducial systems [15], [23], [24], [30].

B. Pattern Detection for Event Cameras

Conventional pattern detection approaches are not directly applicable to event-based data due to its frameless and binary-valued nature [34]. As a straightforward approach, active targets (*e.g.,* LCD or blinking LED) and wand are adopted for event cameras [18], [19], [35]–[38]. Their design is well-tailored to the triggering mechanism of events, making them

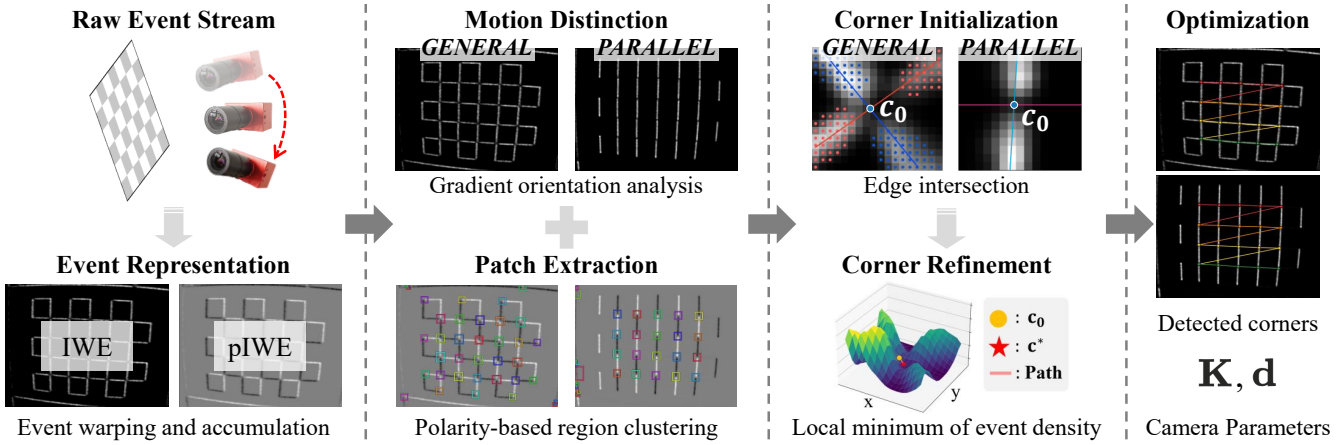


Fig. 2. Overview of the proposed calibration pipeline. The framework converts a raw event stream into both IWE and pIWE, which are representations well-suited for extracting corner points. These representations are then used by the motion distinction module to determine the motion case and by the pattern detection module to detect corners accordingly. The detected corners are then fed into the intrinsic optimization process to estimate camera parameters $\mathbf{K}, \mathbf{d}$.

intuitive and accurate. However, these approaches require specialized hardware and corresponding methodologies.

In the other directions, several studies attempt to detect circular or circle-grid patterns directly from a raw event stream [20]–[22]. These patterns offer robustness, as events are generated along every edge regardless of motion direction. In these approaches, events are typically clustered and modeled by fitting cylindrical or elliptical shapes to a raw event stream. They share the high sensitivity of circular patterns to lens distortion, with cylindrical fitting making precise localization even more difficult. More recently, Wu *et al.* [39] propose a Hough Transform-based approach with a two-stage circle center localization strategy. However, its localization accuracy remains sensitive to temporal window duration.

With the recent advances in learning-based approaches, some methods [16], [17] leverage the event-to-video reconstruction network [40] to convert a raw event stream into grayscale images and apply the Harris corner detector [41] to identify fiducial points. However, event-to-video reconstruction introduces artifacts [42], which may degrade the accuracy of subsequent corner localization. These limitations motivate direct detection of fiducial points from event-domain representations rather than reconstructed images.

In this work, we detect fiducial points directly from event representation, specifically IWE [43], rather than relying on imprecise reconstruction. To this end, we analyze event generation at corner points and propose a calibration pipeline specialized to event characteristics.

III. Problem Statement

We investigate the event camera calibration directly in the event domain, without relying on grayscale image reconstruction. Calibrating event cameras with a checkerboard is challenging due to asynchronous event generation, which leads to blurred edges when events are accumulated [25]. In addition, the absence of events at the corner intersection makes corner detection difficult. Furthermore, when the camera motion is approximately parallel to checkerboard edge, its event response is suppressed, limiting the observable corner structure. To address these, we propose an event camera calibration pipeline using a checkerboard pattern (see Fig. 2).

Our goal is to estimate camera parameters, accounting for event-specific characteristics. Our framework is built upon the event characteristics of checkerboard corners and edges analyzed in Sec. IV. We take as input a raw event stream $\mathcal{E} = \{e_k\}_{k=1}^{N}$, where each event $e_k = (\mathbf{x}_k, t_k, p_k)$ consists of the pixel location $\mathbf{x}_k$, timestamp t_k, and polarity p_k (-1 or $+1$). We then convert the raw event stream into IWE and pIWE through event warping and accumulation to obtain motion-compensated event representations (Sec. V). Using the gradient orientation distribution of IWE, we distinguish the motion case for each event window and select the corresponding corner initialization strategy (Sec. VI). Given the resulting motion case, we initialize checkerboard corners by leveraging the corresponding edge cues and then refine them toward the local minimum of event density to obtain subpixel accuracy. The refined corners are fed into the batch optimization process to estimate the camera parameters $\mathbf{K}$ and $\mathbf{d}$ (Sec. VII). Moreover, we extend our framework to AprilTag detection [15], enabling reliable tag identification even under partial occlusion (Sec. VIII), and to stereo event camera calibration using synchronized event cameras (Sec. IX).

IV. Event Rate on Checkerboard Corner

Understanding how events are generated around an ideal checkerboard corner helps to explain the underlying characteristics of event data and to design more robust detection methods. Here, we analyze that events are not triggered in the checkerboard corner, using the intensity model of the checkerboard corner in frame-based cameras [29] and the event generation rate formulation [44].

A. Intensity Model on Checkerboard Corner

The ideal intensity distribution of the checkerboard corner can be defined with a step function $\rho(\cdot)$ as [29]:

$$\hat{I}(\mathbf{x}) = \rho(\mathbf{n}_1^\top \mathbf{x})\rho(\mathbf{n}_2^\top \mathbf{x}),\ \ \rho(x) = \begin{cases} +1 & \text{if } x > 0, \\ -1 & \text{otherwise,} \end{cases} \tag{1}$$

where $\mathbf{n}_i$ denotes unit normals to checkerboard edges. Using this definition, Krüger *et al.* [29] model the checkerboard corner intensity $\tilde{I}$ by replacing $\rho(\cdot)$ with a centered sigmoid smoothed by a Gaussian of width σ:

$$\tilde{I}(\mathbf{x}) \approx S(\mathbf{n}_1^\top \mathbf{x}, \sigma) S(\mathbf{n}_2^\top \mathbf{x}, \sigma), \tag{2}$$

where $S(x,\sigma) = 2/(1+e^{-x/\sigma}) - 1$ is a centered sigmoid. In this model, checkerboard corner is located at $\mathbf{x} = \mathbf{0}$. Note that since $S(x,\sigma) \in (-1,1)$ by construction, $\tilde{I}(\mathbf{x}) \in [-1,1]$ represents the idealized contrast pattern of the checkerboard, rather than the actual brightness observed by the camera.

B. Event Generation Rate

In this subsection, we review the relationship between Event Rate (ER) and the gradient derived by Lin *et al.* [44]. Based on the event-triggering mechanism, an event is generated whenever the change in logarithmic intensity at a pixel exceeds a fixed contrast threshold [2]:

$$\Delta L(\mathbf{x_k}, t_k) := L(\mathbf{x}_k, t_k) - L(\mathbf{x}_k, t_k - \Delta t_k) = p_k C, \tag{3}$$

where $L(\mathbf{x}, t) = \log I(\mathbf{x}, t)$ denotes the logarithmic image intensity, $C > 0$ is the constant threshold, and Δt_k represents the duration since the pixel last generated an event.

To analyze ER, it is necessary to first characterize how events accumulate over time interval. Since the rate is defined in terms of the number of events generated within a temporal window, it is necessary to relate the accumulated events to the underlying logarithmic intensity signal that drives event generation. Based on Eq. (3), an event e_k can be written in distributional form as [45]:

$$e_k(\mathbf{x}, t) = p_k C\, \delta(t - t_k)\, \delta(\mathbf{x} - \mathbf{x}_k), \tag{4}$$

where $\delta(\cdot)$ indicates the Dirac delta function. Over a time interval of length t, the events accumulated at pixel $\mathbf{x}$ can be written as:

$$\int_0^t \sum_k e_k(\mathbf{x}, \tau)\, d\tau = L(\mathbf{x}, t) - L(\mathbf{x}, 0) + \int_0^t \eta(\mathbf{x}, \tau)\, d\tau, \tag{5}$$

where $\eta(\mathbf{x}, t)$ denotes sensor noise [2].

From the event models described in Eq. (4) and Eq. (5), Lin *et al.* [44] define ER as follows:

$$R_e(\mathbf{x}, t, \Delta t) = \frac{\int_{t-\Delta t/2}^{t+\Delta t/2} \sum_k \delta(\tau - t_k)\delta(\mathbf{x} - \mathbf{x}_k) d\tau}{\Delta t}, \tag{6}$$

where $R_e(\mathbf{x}, t, \Delta t)$ is ER within short time interval $[t - \Delta t/2,\ t + \Delta t/2]$, which indicates the average number of events accumulated within a short time interval of duration Δt. In addition, they formulate the intensity changing rate within same time interval, omitting noise term in Eq. (5):

$$\begin{aligned} \frac{\Delta L(\mathbf{x}, t)}{\Delta t} &= \frac{L(\mathbf{x}, t + \Delta t/2) - L(\mathbf{x}, t - \Delta t/2)}{\Delta t} \\ &= \frac{\int_{t-\Delta t/2}^{t+\Delta t/2} \sum_k e_k(\mathbf{x}, \tau) d\tau}{\Delta t}. \end{aligned} \tag{7}$$

By taking the magnitude of Eq. (7) and substituting the event model $e_k(\mathbf{x}, t)$ using Eq. (4), the resulting intensity changing rate magnitude can be approximated by the event rate scaled by the constant threshold C:

$$\begin{aligned} \left| \frac{\Delta L(\mathbf{x}, t)}{\Delta t} \right| &\approx \frac{C \int_{t-\Delta t/2}^{t+\Delta t/2} \sum_k \delta(\tau - t_k)\delta(\mathbf{x} - \mathbf{x}_k) d\tau}{\Delta t} \\ &= C \cdot R_e(\mathbf{x}, t, \Delta t), \end{aligned} \tag{8}$$

where the polarity p_k vanishes since the magnitude operator eliminates its sign.

Using the brightness constancy assumption, the intensity changing rate also can be approximated as [46]:

$$\frac{\Delta L(\mathbf{x}, t)}{\Delta t} \approx -\nabla L(\mathbf{x}, t) \cdot \mathbf{v}, \tag{9}$$

where $\mathbf{v}$ is the constant motion vector on image plane [46]. Finally, combining Eq. (8) and Eq. (9), Lin *et al.* [44] establish a relationship between ER and the gradient as follows:

$$R_e(\mathbf{x}, t, \Delta t) \approx \frac{1}{C} |\nabla L(\mathbf{x}, t) \cdot \mathbf{v}|. \tag{10}$$

For simplicity, we omit the temporal variable t in the following analysis, using $L(\mathbf{x})$ and $R_e(\mathbf{x})$ instead of $L(\mathbf{x}, t)$ and $R_e(\mathbf{x}, t, \Delta t)$, respectively.

C. Event Rate on Checkerboard Corner

To analyze ER on checkerboard corner, we combine Eq. (2) and Eq. (10). Before combining them, we first reformulate centered sigmoid $S(x,\sigma)$, noting that $S(x,\sigma)$ can be expressed as a hyperbolic tangent function,

$$\begin{aligned} S(x,\sigma) &= \frac{2}{1 + e^{-x/\sigma}} - 1 \\ &= \frac{1 - e^{-x/\sigma}}{1 + e^{-x/\sigma}} \\ &= \frac{e^{x/2\sigma} - e^{-x/2\sigma}}{e^{x/2\sigma} + e^{-x/2\sigma}} \\ &= \tanh\left(\frac{x}{2\sigma}\right). \end{aligned} \tag{11}$$

Then, we recall that the hyperbolic tangent function has the following Taylor expansion:

$$\tanh(x) = x - \frac{1}{3}x^3 + \frac{2}{15}x^5 - \cdots, \ \text{when } |x| < \frac{\pi}{2}. \tag{12}$$

Since we are primarily interested in the neighborhood of the ideal checkerboard corner, where its region is in $|x| \ll 1$, the Taylor expansion of $\tanh(x)$ is valid, and the higher-order terms can be neglected as

$$S(x,\sigma) = \tanh\left(\frac{x}{2\sigma}\right) \approx \frac{x}{2\sigma} + \mathcal{O}(x^3) \approx \frac{x}{2\sigma}, \tag{13}$$

where $\mathcal{O}(x^3)$ denotes the third-order remainder.

Based on this convention, we reformulate Eq. (2) using the approximation in Eq. (13), deriving its local analytical form around the checkerboard corner:

$$\tilde{I}(\mathbf{x}) \approx \frac{1}{4\sigma^2}(\mathbf{n}_1^\top \mathbf{x})(\mathbf{n}_2^\top \mathbf{x}), \quad \nabla \tilde{I}(\mathbf{x}) \approx G\,\mathbf{x}, \tag{14}$$

where $G = \frac{1}{4\sigma^2}(\mathbf{n}_1\mathbf{n}_2^\top + \mathbf{n}_2\mathbf{n}_1^\top)$ is symmetric. Since $\tilde{I}(\mathbf{x})$ represents an idealized contrast pattern rather than an actual

brightness value, we obtain the image intensity via a linear camera response applied to $\tilde{I}(\mathbf{x})$ [29]:

$$I(\mathbf{x}) = I_0 + \alpha\,\tilde{I}(\mathbf{x}), \tag{15}$$

where $\alpha > 0$ and $I_0 > 0$ denote the camera gain and offset, respectively. Through this response, $\tilde{I} \in [-1, 1]$ is mapped to the image intensity I within the dynamic range of camera.

Having established the local intensity model of the checkerboard corner, we now analyze the event rate on checkerboard corner by substituting Eq. (15) into the event rate formulation in Eq. (10). Since $L(\mathbf{x}) = \log I(\mathbf{x})$, we have

$$\begin{aligned}\nabla L(\mathbf{x}) &= \frac{\nabla I(\mathbf{x})}{I(\mathbf{x})} \\ &= \frac{\alpha}{I_0 + \alpha\,\tilde{I}(\mathbf{x})}\nabla\tilde{I}(\mathbf{x}) \\ &\approx \frac{\alpha}{I_0 + \alpha\,\tilde{I}(\mathbf{x})} G\mathbf{x}.\end{aligned} \tag{16}$$

Since $\tilde{I}(\mathbf{x}) \approx 0$ around the corner, the term $\alpha/(I_0 + \alpha\,\tilde{I}(\mathbf{x}))$ can be approximated by α/I_0 and treated as constant. Therefore, Eq. (10) can be rewritten as

$$\begin{aligned}R_e(\mathbf{x}) &\approx \frac{1}{C}\left|\nabla L(\mathbf{x}) \cdot \mathbf{v}\right| \\ &\approx \gamma\left|(G\mathbf{x}) \cdot \mathbf{v}\right| = \gamma\left|(G\mathbf{v})^\top \mathbf{x}\right|,\end{aligned} \tag{17}$$

where $\gamma = \alpha/CI_0$ is a constant factor.

To interpret the meaning of Eq. (17), we first investigate whether G can be zero. Note that, in any valid checkerboard structure, the two edge normals cannot be parallel and are therefore linearly independent. With this geometric constraint, we examine whether G can vanish by multiplying G by $\mathbf{n}_1$:

$$\begin{aligned}G\mathbf{n}_1 &= \frac{1}{4\sigma^2}(\mathbf{n}_1\mathbf{n}_2^\top + \mathbf{n}_2\mathbf{n}_1^\top)\mathbf{n}_1 \\ &= \frac{1}{4\sigma^2}\Big[(\mathbf{n}_1\mathbf{n}_2^\top)\mathbf{n}_1 + (\mathbf{n}_2\mathbf{n}_1^\top)\mathbf{n}_1\Big] \\ &= \frac{1}{4\sigma^2}\Big[(\mathbf{n}_2^\top\mathbf{n}_1)\mathbf{n}_1 + \mathbf{n}_2\Big].\end{aligned} \tag{18}$$

If G is zero, then $G\mathbf{n}_1 = \mathbf{0}$ must hold, which means that the following equation holds:

$$(\mathbf{n}_2^\top\mathbf{n}_1)\mathbf{n}_1 + \mathbf{n}_2 = \mathbf{0}. \tag{19}$$

Thus, $\mathbf{n}_2$ can be written as

$$\begin{aligned}\mathbf{n}_2 &= -(\mathbf{n}_2^\top\mathbf{n}_1)\mathbf{n}_1 \\ &= \lambda\mathbf{n}_1, \quad \text{for some scalar } \lambda.\end{aligned} \tag{20}$$

This implies that $\mathbf{n}_1$ and $\mathbf{n}_2$ are linearly dependent, which directly contradicts the valid checkerboard structure that requires the two edge normals to be linearly independent. Hence, G cannot be zero for checkerboard structure.

Since G is nonzero for a valid checkerboard corner, ER becomes zero at the exact corner $\mathbf{x} = \mathbf{0}$ or in the degenerate case $\mathbf{v} = \mathbf{0}$. Thus, from Eq. (17), we find that

$$\mathbf{x} = \mathbf{0} \implies R_e(\mathbf{0}) = 0. \tag{21}$$

This result holds locally near a checkerboard corner, where the image gradients are modeled as a pair of intersecting edges.

In contrast, the intensity model of the checkerboard edge can be modeled as

$$\tilde{I}(\mathbf{x}) \approx \frac{1}{2\sigma}(\mathbf{n}^\top\mathbf{x}), \quad \nabla\tilde{I}(\mathbf{x}) \approx \frac{\mathbf{n}}{2\sigma}. \tag{22}$$

Here, the checkerboard edge corresponds to the set of points $\mathbf{x}$ that satisfy $\mathbf{n}^\top\mathbf{x} = \mathbf{0}$. Then we can calculate $\nabla L(\mathbf{x})$ in edge case as follows:

$$\nabla L(\mathbf{x}) \approx \frac{\alpha}{I_0 + \alpha\,\tilde{I}(\mathbf{x})}\,\frac{\mathbf{n}}{2\sigma}. \tag{23}$$

As in the checkerboard corner case, since $\tilde{I}(\mathbf{x}) \approx 0$ near the edge, we can formulate ER on the checkerboard edge as follows:

$$\begin{aligned}R_e(\mathbf{x}) &\approx \frac{1}{C}\left|\nabla L(\mathbf{x}) \cdot \mathbf{v}\right| \\ &\approx \kappa\left|\mathbf{n} \cdot \mathbf{v}\right|, \quad \text{where } \kappa = \gamma/2\sigma.\end{aligned} \tag{24}$$

This expression indicates that events are reliably generated along the edge unless the motion vector is orthogonal to the edge normal. Conversely, when the motion is parallel to the edge direction, no events are produced. In other words, events may be triggered only on one side of the checkerboard edge under such motion [22].

While Eq. (17) uses the absolute value to describe the event rate, the quantity $(G\mathbf{v})^\top\mathbf{x}$ retains the polarity information. Using the definition of G in Eq. (14), under the parallel motion condition $\mathbf{n}_2^\top\mathbf{v} \approx 0$, we obtain

$$\begin{aligned}(G\mathbf{v})^\top\mathbf{x} &= \frac{1}{4\sigma^2}\left[(\mathbf{n}_2^\top\mathbf{v})(\mathbf{n}_1^\top\mathbf{x}) + (\mathbf{n}_1^\top\mathbf{v})(\mathbf{n}_2^\top\mathbf{x})\right] \\ &\approx \frac{1}{4\sigma^2}(\mathbf{n}_1^\top\mathbf{v})(\mathbf{n}_2^\top\mathbf{x}).\end{aligned} \tag{25}$$

Within an event window, $\mathbf{n}_1^\top\mathbf{v}$ remains constant, while $\mathbf{n}_2^\top\mathbf{x}$ takes opposite signs on the two sides of the corresponding edge. Accordingly, the sign of $(G\mathbf{v})^\top\mathbf{x}$ changes across this boundary, yielding opposite event polarities on its two sides. Thus, although ER of this edge is suppressed under parallel motion, its position can still be inferred from the remaining polarity transition near the corner.

In conclusion, under the first-order local model, checkerboard corners exhibit near-zero event density, as shown in Eq. (21), while Eq. (24) shows that edge responses depend on the motion direction. Under parallel motion, one edge response is suppressed, while the polarity transition across the suppressed edge remains available for localization, as shown in Eq. (25). These complementary properties motivate our detection strategy, where edge cues support initialization, polarity transition enables suppressed edge recovery, and low event density at corner guides refinement.

V. Event Representation

Event cameras capture visual information asynchronously. This asynchronous stream forms spatio-temporal trajectories based on relative scene motion or ego-motion [25], [43]. Thus, simple accumulation of these events on a static image plane causes motion blur. To mitigate it, various event representations have been proposed [25], [43], [47]–[49]. Among them, we adopt IWE [25], [43], which warps events to a reference

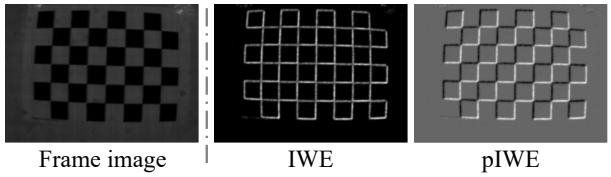


Fig. 3. Event representation. IWE accumulates events regardless of polarity, whereas pIWE preserves polarity information. Both representations are rescaled to $[0, 255]$ for visualization only.

time under a motion model to enforce geometric alignment and suppress motion blur. In contrast to time-surfaces [47], [48] that encode temporal recency at each pixel, IWE aligns spatio-temporal event trajectories onto a common instant, yielding an edge image consistent with the single-instant assumption of camera calibration [27].

To construct IWE, we first divide the input event stream $\mathcal{E}$ into M coherent windows with fixed time d_t, $\mathcal{E} = \bigcup_{i=1}^{M} \mathcal{E}_i$, and process each window $\mathcal{E}_i$ independently. For each window, we warp each event to reference time t_{ref}, using the warping function [25]:

$$\begin{aligned} e_k = (\mathbf{x}_k, t_k, p_k) \mapsto e'_k = (\mathbf{x}'_k, t_{\text{ref}}, p_k), \\ \mathbf{x}'_k = \mathbf{x}_k + (t_{\text{ref}} - t_k)\,\mathbf{v}, \end{aligned} \tag{26}$$

where $\mathbf{v} \in \mathbb{R}^2$ denotes the constant image-plane motion vector and $\mathbf{x}'_k$ is the pixel location of the warped event e'_k. This model assumes linear camera motion within each window, which is a valid assumption in calibration scenarios where data can be collected under a controlled environment [27].

Then, we build Image of Warped Events (IWE):

$$H(\mathbf{x}; \mathbf{v}) = \sum_{k=1}^{N_e} p_k \delta(\mathbf{x} - \mathbf{x}'_k), \tag{27}$$

where N_e represents the number of events in each window, each pixel $\mathbf{x}$ accumulates the number of warped events whose locations $\mathbf{x}'_k$ spatially coincide with it, and $\delta(\cdot)$ indicates the Dirac delta function. To estimate the optimal motion vector $\mathbf{v}^*$, we maximize the variance of IWE:

$$\mathbf{v}^* = \arg\max_{\mathbf{v} \in \mathbb{R}^2} \operatorname{Var}\big(H(\mathbf{x}; \mathbf{v})\big). \tag{28}$$

This objective encourages sharper and more coherent event alignment [25], [43].

Using the estimated motion vector $\mathbf{v}^*$, we can construct IWE and polarity-aware IWE (pIWE) for each window. IWE $H(\mathbf{x}; \mathbf{v}^*)$ aggregates events regardless of their polarity by setting $p_k = 1$ for all events. It stably captures the landscape of event density without being affected by polarity cancellation. In contrast, pIWE $H_p(\mathbf{x}; \mathbf{v}^*)$ holds the polarity p_k of each event. Using pIWE is effective for identifying black–white alternation of checkerboard patterns, as it preserves polarity contrast. We use these event representations in the subsequent corner detection pipeline (see Fig. 3).

We note that, unlike grayscale images from the reconstruction network [40], which estimates absolute scene brightness, IWE and pIWE are motion-compensated event representations that retain event density. Consequently, ER is directly compatible with these representations.

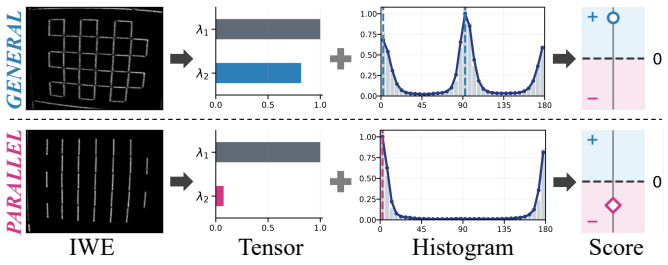


Fig. 4. Motion distinction procedure. The top and bottom rows show *GENERAL* and *PARALLEL* cases, respectively. The gradient orientation distribution of each IWE is summarized by tensor and histogram statistics, which are combined into the final motion distinction score.

VI. Gradient-Based Motion Distinction

As discussed in Sec. IV-C, the visibility of checkerboard edges depends on the image-plane motion direction. Based on this observation, we consider two motion cases. We refer to the case in which both checkerboard edges remain sufficiently visible as *GENERAL*. In contrast, when the image-plane motion is approximately parallel to one checkerboard edge, its event response is suppressed while the other edge remains dominant. We refer to this case as *PARALLEL*.

In our previous work [26], *PARALLEL* case is filtered out during corner initialization because it does not satisfy the four-cluster configuration required by the propagation procedure. In this work, we instead distinguish motion from the gradient orientation distribution of IWE and apply the corresponding corner initialization strategy. As illustrated in Fig. 4, *GENERAL* case produces two dominant gradient orientations, whereas *PARALLEL* case produces a single dominant orientation. We quantify this difference to determine the motion case for each event window.

A. Gradient Orientation Statistics

We first smooth IWE H with a Gaussian kernel, yielding $\bar{H}$. At each image location $\mathbf{x}_i$, we compute the gradient magnitude and normalized direction as

$$w_i^{(g)} = \left\|\nabla \bar{H}(\mathbf{x}_i)\right\|_2, \qquad \mathbf{u}_i = \frac{\nabla \bar{H}(\mathbf{x}_i)}{\left\|\nabla \bar{H}(\mathbf{x}_i)\right\|_2}. \tag{29}$$

We use $w_i^{(g)}$ as the weight so that stronger gradient responses contribute more to the orientation statistics.

To characterize the gradient orientation distribution, we define two statistics: the tensor ratio r_{T} and the histogram ratio r_{H}. To compute tensor ratio r_{T}, we construct a weighted second-order orientation tensor [41], [50] as

$$\mathbf{M}_g = \frac{1}{W_g} \sum_i w_i^{(g)} \mathbf{u}_i \mathbf{u}_i^\top, \qquad W_g = \sum_i w_i^{(g)}. \tag{30}$$

Then, we define the tensor ratio as

$$r_{\text{T}} = \frac{\lambda_2}{\max(\lambda_1, \varepsilon)}, \tag{31}$$

where $\lambda_1 \geq \lambda_2 \geq 0$ denote the eigenvalues of $\mathbf{M}_g$ and ε is a small constant for numerical stability. A dominant single orientation makes $\mathbf{M}_g$ approximately rank one, resulting in a

small λ_2 and thus a low r_{T}. In contrast, when gradients are distributed over two distinct orientations, the gap between λ_1 and λ_2 decreases, resulting in a higher r_{T}.

The tensor ratio r_{T} describes the global spread of gradient orientations, but it does not explicitly indicate whether a distinct secondary orientation exists. We therefore construct a magnitude-weighted axial orientation histogram to derive the histogram ratio r_{H}. To construct the histogram, we accumulate the weights within each angular bin:

$$h_k = \sum_{i:\theta_i \in \mathcal{B}_k} w_i^{(g)}. \tag{32}$$

Here, θ_i denotes the orientation of $\mathbf{u}_i$ over $[0, \pi)$, $\mathcal{B}_k$ is the k-th angular bin, and h_k is its accumulated support. To quantify the presence of a distinct secondary orientation, we compare the dominant orientation peak with the strongest sufficiently separated secondary peak. The histogram ratio is defined as

$$r_{\mathrm{H}} = \frac{h_{\mathrm{sec}}}{\max(h_{\mathrm{pri}}, \varepsilon)}, \tag{33}$$

where h_{pri} and h_{sec} denote the accumulated supports of the dominant and secondary peaks, respectively. If no valid secondary peak exists, r_{H} is set to zero.

In conclusion, r_{T} and r_{H} provide complementary measures of the gradient orientation distribution. As shown in Fig. 4, *GENERAL* case typically yields higher values of both r_{T} and r_{H} due to the presence of two dominant orientations, whereas *PARALLEL* case yields lower values because a single orientation becomes dominant.

B. Motion Distinction Score

Since the tensor ratio r_{T} and histogram ratio r_{H} characterize different aspects of gradient orientation distribution, we normalize them to common scale before combination [51]. For each statistic, we define midpoint m and scaling factor β using empirically selected reference anchors:

$$\begin{aligned} m_{\mathrm{T}} &= \frac{r_{\mathrm{T,G}} + r_{\mathrm{T,P}}}{2}, \quad \beta_{\mathrm{T}} = \frac{r_{\mathrm{T,G}} - r_{\mathrm{T,P}}}{2}, \\ m_{\mathrm{H}} &= \frac{r_{\mathrm{H,G}} + r_{\mathrm{H,P}}}{2}, \quad \beta_{\mathrm{H}} = \frac{r_{\mathrm{H,G}} - r_{\mathrm{H,P}}}{2}. \end{aligned} \tag{34}$$

where $r_{\mathrm{T,G}}$ and $r_{\mathrm{T,P}}$ denote the tensor ratio anchors for *GENERAL* and *PARALLEL* cases, and $r_{\mathrm{H,G}}$ and $r_{\mathrm{H,P}}$ denote the corresponding histogram ratio anchors.

Using these, the normalized statistics are defined as

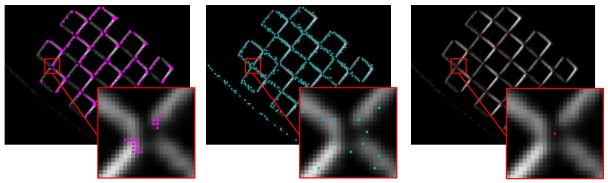
$$z_{\mathrm{T}} = \frac{r_{\mathrm{T}} - m_{\mathrm{T}}}{\beta_{\mathrm{T}}}, \qquad z_{\mathrm{H}} = \frac{r_{\mathrm{H}} - m_{\mathrm{H}}}{\beta_{\mathrm{H}}}. \tag{35}$$

This normalization places *PARALLEL* and *GENERAL* reference anchors at -1 and $+1$, respectively. Thus, negative normalized values support *PARALLEL* case, whereas positive values support *GENERAL* case. We then combine the two statistics into a motion distinction score,

$$\Phi = \frac{w_T z_T + w_H z_H}{w_T + w_H}, \tag{36}$$

where w_T and w_H control the relative contributions of the tensor and histogram statistics, respectively. The motion case is determined by the sign of Φ:

$$\Phi \geq 0 \Rightarrow \textit{GENERAL}, \quad \Phi < 0 \Rightarrow \textit{PARALLEL}. \tag{37}$$



(a) Harris (b) Shi-Tomasi (c) *Ours*

Fig. 5. Comparison with conventional methods. Harris [41] and Shi-Tomasi [52] fail under event data, whereas the proposed method achieves accurate and stable localization.

We then apply the corresponding corner initialization strategy for each motion case. Initialization procedures for *GENERAL* and *PARALLEL* cases are described in Secs. VII-B and VII-C.

VII. Corner Detection and Calibration

Here, we propose our calibration framework, including a checkerboard corner detection pipeline built upon our analytical insights from Sec. IV-C. Our framework consists of four steps: (i) patch extraction, (ii) corner initialization, (iii) corner refinement, and (iv) camera parameters optimization. First, we extract local patches around checkerboard corners to isolate candidate regions. Second, for each patch, we initialize corner positions by identifying local patterns consistent with the geometry of the checkerboard corner. Third, we refine the initial corners to sub-pixel accuracy by optimization. Finally, we aggregate the refined corners and use them for calibration.

A. Patch Extraction

To pinpoint the refined corner $\mathbf{c}^*$ precisely, we extract small patches to capture local pattern near each corner. While our overall calibration pipeline primarily utilizes IWE, we leverage polarity contrast of pIWE during the patch extraction to detect initial corners.

To extract each patch in pIWE, we adopt circular boundaries inspired by Bok *et al.* [53]. Since a checkerboard corner lies at the intersection of black and white regions, we select pixels whose circular boundaries contain both positive and negative polarities. Then, the selected pixels close to each other are merged into the same clusters. Each cluster is enclosed by its smallest bounding rectangle within pIWE. Since IWE shares the same pixel coordinates with pIWE, we can extract the corresponding candidate corner regions from IWE. We denote such candidate patches as $\mathcal{P}$.

At first glance, it seems to be possible to apply conventional detection algorithms [41], [52] to IWE. However, even with IWE, corner detection remains highly unreliable on event data (see Fig. 5). As analyzed in Eq. (21), this is because corner regions barely trigger any events.

B. Corner Initialization for General Case

For an event window assigned to *GENERAL* case, the checkerboard edge structure remains sufficiently visible in IWE. Since events rarely occur at the corner itself, as analyzed in Eq. (21), we estimate the initial corner $\mathbf{c}_0$ within each candidate patch $\mathbf{P} \in \mathcal{P}$ from the surrounding edge responses. To

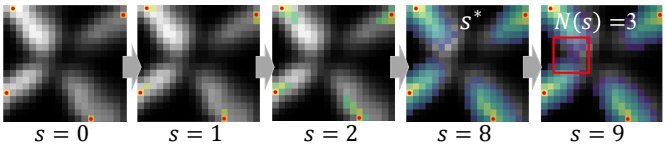


Fig. 6. Pixel propagation from seed pixels. Each red point means seed pixel.

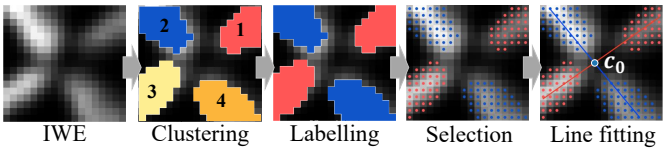


Fig. 7. Corner initialization procedure for *GENERAL* case.

handle this issue, we first leverage edge-based cues to estimate the initial corner $\mathbf{c}_0$. As each pixel value in IWE represents the event density, pixels with high values correspond to edges where distinct motion occurs. Leveraging this property, we can define seed pixels $\mathcal{S}$ as pixels that have higher values than their eight neighboring pixels. Seed pixels $\mathcal{S}$ are highly likely to belong to certain edges, so we detect them from the boundary regions within patch $\mathbf{P}$ (see Fig. 6). The seed pixels $\mathcal{S}$ are defined as follows:

$$\mathcal{S} \;=\; \big\{\mathbf{x}\in\mathbf{P} \;\big|\; H(\mathbf{x}) > H(\mathbf{y}),\ \forall \mathbf{y}\in\mathcal{N}_8(\mathbf{x})\big\}, \tag{38}$$

where $\mathbf{x}$ is a pixel in the patch $\mathbf{P}$ and $\mathcal{N}_8(\mathbf{x})$ represents eight neighbors of $\mathbf{x}$. To initialize precise corner points, we propagate from each seed pixel $\mathfrak{s} \in \mathcal{S}$ toward the second-largest neighbor in $\mathcal{N}_8(\mathfrak{s})$ step by step (see Fig. 6). After each seed pixel $\mathfrak{s}$ propagates one step toward its neighboring pixel, it merges with the propagated pixel, forming a cluster. Let $N(s)$ denote the number of non-empty clusters after merging at step $s \in \mathbb{R}$. Each patch generally maintains $N(s) = 4$, corresponding to the four edges of the checkerboard pattern. When $N(s) \neq 4$, it indicates either a single-edge response or a misdetected seed, and such cases are filtered out. As propagation continues, clusters gradually merge, decreasing $N(s)$. We therefore define s^* as the largest propagation step for which $N(s) = 4$:

$$s^* \;=\; \max\big\{\, s\in\mathbb{R} \;\big|\; N(s) = 4 \,\big\}. \tag{39}$$

At step s^*, we order the four clusters counterclockwise by label $l \in \{1, 2, 3, 4\}$. We then define two lines based on index parity: ℓ_{odd} is the line fitted to all pixels belonging to clusters with an odd label, and ℓ_{even} is the line fitted to all pixels belonging to clusters with an even label. Then, we determine the corner $\mathbf{c}_0$ as the intersection $\mathbf{c}_0 = \ell_{\text{odd}} \times \ell_{\text{even}}$ (see Fig. 7). We can filter out the invalid patch if the obtained $\mathbf{c}_0$ is not in $\mathbf{P}$. This approach enables us to initialize a reliable corner $\mathbf{c}_0$ when few or no events occur around the actual corner.

C. Corner Initialization for Parallel Case

For an event window assigned to *PARALLEL* case, one checkerboard edge becomes unobservable. We therefore recover the suppressed edge based on the polarity transition, as analyzed in Eq. (25). Then, we combine it with the remaining visible edge for corner initialization. As illustrated in Fig. 8, the procedure consists of suppressed edge recovery, visible edge fitting, and corner initialization.

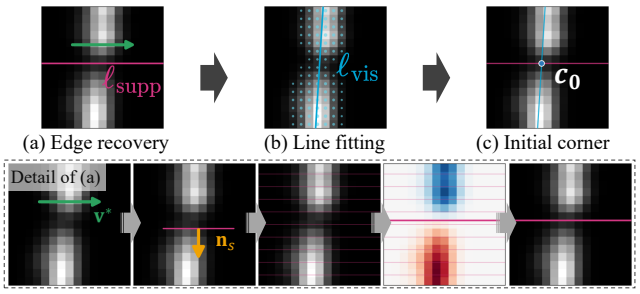


Fig. 8. Corner initialization procedure for *PARALLEL* case.

Under the parallel motion condition, the motion is orthogonal to the normal of suppressed edge:

$$\mathbf{n}_s^\top \mathbf{v}^* = 0, \tag{40}$$

where $\mathbf{v}^*$ is the estimated motion vector in Sec. V and $\mathbf{n}_s$ is unit normal of suppressed edge ℓ_{supp}. Since we already estimate $\mathbf{v}^*$, we obtain $\mathbf{n}_s$ by rotating motion vector by $90°$.

The motion direction determines the orientation of ℓ_{supp} but not its offset b_s. To estimate this offset, we locate the polarity transition using pIWE H_p. For each non-zero pIWE sample at pixel $\mathbf{y}_j$, we define its projected coordinate, the sign of pIWE, and its magnitude as

$$\zeta_j = \mathbf{n}_s^\top \mathbf{y}_j, \quad \chi_j = \operatorname{sign}(H_p(\mathbf{y}_j)), \quad \omega_j^{(p)} = |H_p(\mathbf{y}_j)|. \tag{41}$$

Here, $\operatorname{sign}(\cdot)$ returns $+1$ for positive values and -1 for negative values. For a candidate offset b, we consider two possible sign orderings across the transition, represented by $\xi \in \{-1, +1\}$. The expected pIWE sign is defined as

$$\hat{\chi}_j(b,\xi) = \begin{cases} -\xi, & \zeta_j \le b, \\ \xi, & \zeta_j > b. \end{cases} \tag{42}$$

We estimate the edge offset and sign ordering by minimizing the magnitude-weighted disagreement between observed and expected pIWE signs:

$$(b_s, \xi^*) = \arg\min_{b,\xi} \frac{\sum_j \omega_j^{(p)} \mathbf{1}\left[\chi_j \neq \hat{\chi}_j(b,\xi)\right]}{\sum_j \omega_j^{(p)}}. \tag{43}$$

We then recover the suppressed edge as

$$\ell_{\text{supp}} : \mathbf{n}_s^\top \mathbf{x} = b_s. \tag{44}$$

We next fit the remaining visible edge ℓ_{vis} using its event support and obtain the initial corner $\mathbf{c}_0$ from the intersection of ℓ_{vis} and ℓ_{supp}. By recovering the suppressed edge based on polarity transitions, this approach enables reliable corner initialization even when only one checkerboard edge is visible.

D. Corner Refinement

To obtain the accurate refined corner $\mathbf{c}^*$, we refine the initial corner $\mathbf{c}_0$. Based on our analysis in Eq. (21), we assume that the ideal corner corresponds to the minimum value in

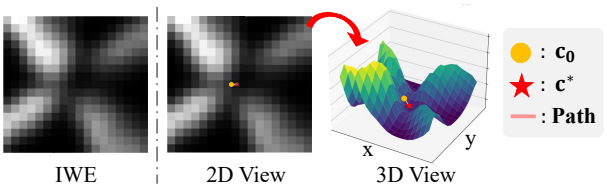


Fig. 9. Visualization of corner refinement on the IWE surface. Its height represents the count of accumulated events. We show the optimization of an initial corner $\mathbf{c}_0$ to its refined corner $\mathbf{c}^*$ along the gradient path.

IWE. Thus, we apply gradient descent to $\mathbf{c}_0$ to find this local minimum (see Fig. 9). During this process, non-corner regions in IWE (*e.g.,* checkerboard squares) may also yield near-zero responses, potentially leading to incorrect convergence. To mitigate this effect, we constrain the optimization using the two edge lines. Let ℓ_1 and ℓ_2 denote the two edge lines from each initializer, *i.e.* ℓ_{odd}, ℓ_{even} for *GENERAL* case and ℓ_{supp}, ℓ_{vis} for *PARALLEL* case. We restrict the refined corner $\mathbf{c}^*$ to the intersection of narrow bands around these lines:

$$\mathbf{c}^* = \arg\min_{\mathbf{c}} H(\mathbf{c}) \quad \text{s.t. } \operatorname{dist}(\mathbf{c}, \ell_i) \le \epsilon, \ i = 1, 2, \tag{45}$$

where $\operatorname{dist}(\mathbf{c}, \ell)$ denotes the Euclidean distance from point $\mathbf{c}$ to line ℓ in the image plane, and ϵ is a tolerance in pixels.

This refinement process encourages the initial corners to move toward the minimum of the event density within the constrained region, achieving sub-pixel accuracy. Through this process, we obtain geometrically consistent corner positions that align with our theoretical analysis.

E. Camera Parameter Optimization

Finally, we perform a batch optimization to estimate the camera parameters. Each of the event windows yields a set of refined 2D corners $\mathcal{C}_i^{2D}$, where $i \in \{1, \dots, M\}$. Each element in $\mathcal{C}_i^{2D}$ corresponds to a refined corner $\mathbf{c}^*$ obtained in the previous process. We also denote the 3D corner coordinates of the calibration board as $\mathcal{C}^{3D}$, which are predefined and fixed during optimization. The refined corner sets from all event windows are jointly used with the predefined 3D corner coordinates to optimize the intrinsic matrix $\mathbf{K}$ and distortion coefficients $\mathbf{d}$ by minimizing the reprojection error using `calibrateCamera`$(\cdot)$ in OpenCV [13]:

$$\mathbf{K}^*, \mathbf{d}^* = \arg\min_{\mathbf{K},\mathbf{d},\mathbf{T}_i} \sum_{i=1}^{M} \left\| \mathcal{C}_i^{2D} - \pi\left(\mathbf{K}, \mathbf{d}, \mathbf{T}_i, \mathcal{C}^{3D}\right) \right\|^2, \tag{46}$$

where $\mathbf{T}_i$ denotes the extrinsic parameters between camera and checkerboard for the i-th event window, and $\pi(\cdot)$ represents the pinhole projection model including distortion.

VIII. AprilTag Detection

By using the polarity contrast of pIWE (Sec. VII), our method can be readily extended to AprilTag detection [15], [24]. AprilTag encodes unique IDs within its patterns, enabling reliable identification even under partial visibility or occlusion. Thus, we demonstrate an extension built upon the proposed pattern detection to ID-based fiducial marker (see Fig. 10).

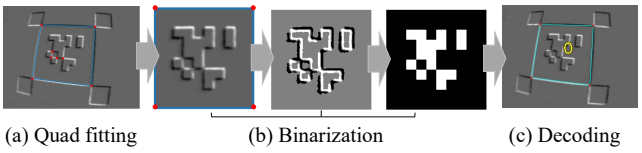


Fig. 10. AprilTag detection procedure.

AprilTag detection consists of three steps: (i) quad fitting, (ii) binarization, and (iii) tag decoding. Under the same assumptions in Sec. III, we capture AprilTag patterns using an event camera. By applying our pattern detection method, we can reliably obtain refined corners $\mathcal{C}_i^{2D}$ corresponding to each divided window. For simplicity, we omit the subscript i in the following description.

A. Quad Fitting

Given the refined corners $\mathcal{C}^{2D}$, we enumerate four-point combinations to obtain quadrilateral candidates $\mathcal{Q}$ (see Fig. 10(a)). Thus, each candidate $\mathcal{Q}$ is defined as a subset of four corners:

$$\mathcal{Q} \subset \mathcal{C}^{2D}, \quad |\mathcal{Q}| = 4,$$

where $|\cdot|$ denotes the number of elements.

We order each subset $\mathcal{Q}$ clockwise and check whether it satisfies the following geometric constraints to verify its validity as a quadrilateral:

- **Angle range constraint:** All internal angles should remain within a valid range.
- **Minimum area threshold:** The quadrilateral must have sufficient area to ensure geometric stability for detection.
- **Edge length consistency:** The quadrilateral should maintain similar side lengths to prevent distorted shapes.

We then remove overlapping quadrilaterals whose enclosing regions largely overlap using non-maximum suppression (NMS). The remaining quads serve as tag candidates for the subsequent process.

B. Binarization

In this process, we normalize each valid quadrilateral $\mathbf{q} \in \mathcal{Q}$ into a grid-like binary pattern to easily identify the tag (see Fig. 10(b)). Based on the four corners of quadrilateral $\mathbf{q}$, we warp the corresponding pIWE region into a normalized square coordinate frame.

From the warped patch, we compute image gradients ∇H_p and project them along motion vector $\mathbf{v}^*$, which is the same in Sec. V, to obtain a contrast map $g = \nabla H_p \cdot \mathbf{v}^*$. This operation emphasizes polarity-aligned structural edges of the tag. We then apply ternary thresholding to g to produce $\{-1, 0, +1\}$ values that represent negative edges, neutral regions, and positive edges. Next, we divide the ternarized patch into an $N \times N$ grid and compute the mean value of each cell to construct a coarse binary structure that encodes the tag pattern. Finally, we binarize the cell using Otsu's adaptive thresholding [54].

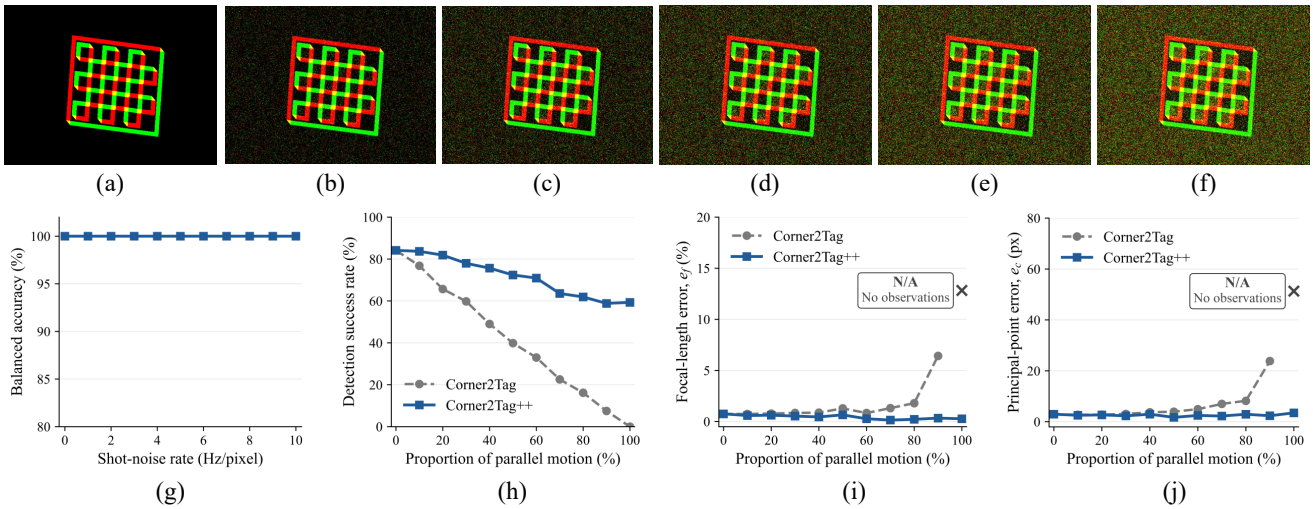


Fig. 11. Synthetic evaluation of the proposed parallel motion handling. (a)-(f) Representative raw-event observations with shot-noise rates of 0, 2, 4, 6, 8, and 10 Hz/pixel, respectively. (g) Balanced accuracy of the motion distinction module under increasing shot-noise rates. (h)-(j) Comparison with [26] as the proportion of parallel motion increases: detection success rate, focal-length error e_f, and principal-point error e_c, respectively.

C. Tag Decoding

To assign the tag ID to the given binary pattern, we pack the binary pattern into an integer codeword and compare it with the predefined AprilTag codebook. We achieve rotation invariance by evaluating the Hamming distance across four possible orientations and select the best match when the minimum Hamming distance falls below a predefined threshold [15]. By doing so, we can obtain the unique tag ID and orientation corresponding to given quadrilateral $\mathbf{q}$.

IX. Stereo Event Camera Calibration

Our framework can be extended to stereo calibration with synchronized event cameras. For each stereo observation, the left and right event streams are selected over the same temporal window with a common reference time t_{ref}. The motion vector is estimated independently for each camera, and the events in each view are warped to t_{ref}. Consequently, the resulting left and right IWE and pIWE represent the checkerboard at the same reference time. The proposed checkerboard detection pipeline is then applied independently to each view.

We retain synchronized observations for which both cameras produce valid checkerboard detections with consistent corner indexing. The common indexing directly establishes the left-right corner correspondences through predefined checkerboard geometry, without requiring additional matching.

The intrinsic parameters of the two cameras are first estimated independently using the camera calibration procedure. With the intrinsic parameters fixed, the relative rotation $\mathbf{R}_{LR}$ and translation $\mathbf{t}_{LR}$ are then estimated using `stereoCalibrate`$(\cdot)$ in OpenCV [13].

X. Experimental Results

In this section, we evaluate the proposed framework on both synthetic and real-world event data. Our method is implemented in C++ and run on a desktop equipped with an Intel Core i7-14700K CPU and 32 GB of RAM. We set the temporal window length d_t to 100 ms for all experiments. For motion distinction, we use 36 orientation histogram bins with a minimum peak separation of 20°, and set $(r_{\text{T},P}, r_{\text{T},G}) = (0.15, 0.40)$ and $(r_{\text{H},P}, r_{\text{H},G}) = (0.20, 0.45)$ with $w_{\text{T}} = w_{\text{H}} = 1$. For corner refinement, we set $\epsilon = 0.5$.

A. Evaluation of Parallel Motion Handling

We evaluate the components introduced to handle parallel motion case through two experiments. The first evaluates the robustness of the motion distinction module to event noise, while the second analyzes checkerboard detection and calibration performance as the proportion of parallel motion observations increases. For the second, we compare with our conference version [26], denoted *Corner2Tag*, which discards observations acquired under parallel motion.

For the first experiment, we inject shot noise into the synthetic event stream at rates from 0 to 10 Hz/pixel while keeping the checkerboard geometry and camera motion fixed. Fig. 11(a)-(f) show representative observations at 0, 2, 4, 6, 8, and 10 Hz/pixel, respectively. We then evaluate the motion distinction module using balanced accuracy over the tested noise levels. As shown in Fig. 11(g), the balanced accuracy remains stable as the shot-noise rate increases, demonstrating robustness to event noise.

For the second experiment, we evaluate our framework, denoted *Corner2Tag++*, as the proportion of parallel motion increases and compare its performance with *Corner2Tag*. While *Corner2Tag* discards observations acquired under parallel motion, *Corner2Tag++* retains them through the newly introduced corner detection method under parallel motion. We evaluate checkerboard detection using the detection success rate and camera calibration accuracy using the focal-length error e_f and principal-point error e_c:

TABLE I
CALIBRATION RESULTS ON SELF-COLLECTED REAL DATA.

Camera	Method	$\sigma(f_x)$	$\sigma(f_y)$	$\sigma(c_x)$	$\sigma(c_y)$	$\sigma(k_1)$	RMSE
DAVIS346	E2Calib [16]	14.34	13.70	13.68	14.54	0.13	1.28
	E2Calib‡	9.73	8.84	11.38	9.39	0.06	0.93
	Ours	**5.51**	**5.25**	**5.91**	**4.85**	**0.03**	**0.64**
DVXplorer	E2Calib [16]	8.17	8.70	12.69	12.22	0.04	1.77
	E2Calib‡	7.20	7.54	8.80	8.04	0.03	1.14
	Ours	**5.21**	**5.30**	**7.94**	**7.91**	**0.02**	**1.10**
EVK4	E2Calib [16]	5.86	5.30	6.80	7.09	**0.01**	1.58
	E2Calib‡	5.21	4.94	5.88	6.08	0.02	1.37
	Ours	**4.31**	**4.07**	**4.49**	**5.04**	**0.01**	**0.82**

‡E2Calib using HyperE2VID [55].

$$e_f = \frac{1}{2}\left(\frac{|f_x - \bar{f}_x|}{\bar{f}_x} + \frac{|f_y - \bar{f}_y|}{\bar{f}_y}\right) \times 100,$$
$$e_c = \frac{1}{2}\left(|c_x - \bar{c}_x| + |c_y - \bar{c}_y|\right), \tag{47}$$

where $\bar{f}_x$, $\bar{f}_y$, $\bar{c}_x$, and $\bar{c}_y$ denote the ground-truth parameters.

As shown in Fig. 11(h), the detection success rate of *Corner2Tag* decreases rapidly as the proportion of parallel observations increases and reaches zero when all observations are acquired under parallel motion. In contrast, *Corner2Tag++* maintains a substantially higher detection success rate as the proportion of parallel motion observations increases.

This difference is also reflected in the calibration results. As illustrated in Fig. 11(i)-(j), *Corner2Tag* exhibits large focal-length and principal-point errors at high parallel motion ratios and eventually fails to provide calibration results because too few valid checkerboard observations remain. In contrast, *Corner2Tag++* maintains comparatively stable calibration errors over the evaluated range. These results show that the newly proposed parallel motion handling maintains checkerboard detection and calibration performance as the proportion of parallel motion increases.

B. Camera Intrinsic Calibration

We evaluate our framework on both real-world and synthetic data. For real-world data, where ground-truth camera parameters are unavailable, we assess calibration repeatability across repeated trials using standard deviation. For synthetic data, where the ground-truth parameters are known, we directly evaluate calibration accuracy.

1) Real-World Experimental Setup: For the real-world experiments, we use three event cameras with different spatial resolutions: DAVIS346 (346×260), DVXplorer (640×480), and Prophesee EVK4 (1280×720). The same 5×6 checkerboard (35 mm) is used for all three cameras. We compare our method with the reconstruction-based E2Calib [16], which uses E2VID [40] to reconstruct grayscale images followed by standard image-based calibration. To account for advances in event-to-image reconstruction, we additionally replace E2VID with the more recent HyperE2VID [55] while keeping the remaining calibration pipeline unchanged. For all checkerboard-based methods, the detected corners are passed to the same OpenCV calibration backend for camera parameter estimation.

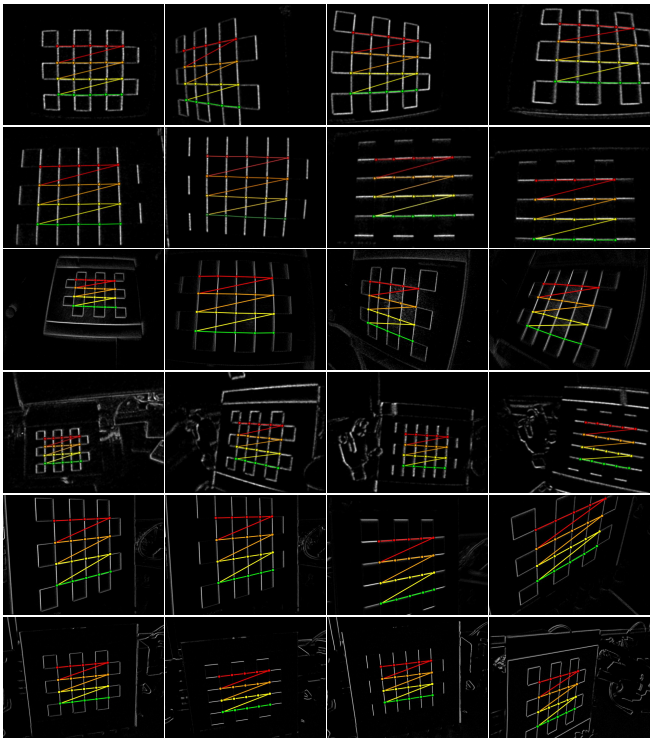


Fig. 12. Checkerboard detection results on real-world data from DAVIS346 (rows 1-2), DVXplorer (3-4), and Prophesee EVK4 (5-6).

Circle-grid-based methods such as EventCalib [20], E-Calib [22] and eKalibr [21] are not included in the real-world comparison. A direct comparison would require separate acquisitions using different physical calibration targets, resulting in different camera–target pose distributions between the checkerboard and circle-grid experiments and making a fair comparison difficult.

2) Calibration Repeatability on Real Data: We evaluate calibration repeatability using the standard deviation of the estimated camera parameters over repeated calibration trials. For each camera, we use only the observations in which all compared methods successfully detect the complete checkerboard. We randomly sample 40 observations without replacement from this common set and use the same subset for all methods. We repeat this procedure 100 times with independently sampled subsets. We use the resulting standard deviations to measure the repeatability of the estimated camera parameters across different calibration observation subsets. We additionally report the reprojection RMSE computed using the mean camera parameters.

As shown in Table I, our method consistently achieves lowest standard deviations for all three event cameras, demonstrating reliable calibration across different sensor configurations. Qualitative corner detection results under diverse viewing conditions are shown in Fig. 12.

3) Synthetic Experimental Setup: For the synthetic experiments, we generate calibration sequences in Blender and simulate the corresponding event streams using v2e [56]. To ensure a fair comparison between checkerboard- and circle-grid-based calibration methods, we use the same planar target

TABLE II
CALIBRATION RESULTS ON SYNTHETIC DATA.

Method	f_x	f_y	c_x	c_y	k_1
GT	500	500	320	240	−0.20
EventCalib* [20]	515.40 ±8.46	514.42 ±8.30	317.42 ±3.06	238.37 ±4.26	– –
E-Calib* [22]	498.67 ±11.54	**498.84** ±11.33	314.99 ±12.33	242.48 ±5.94	−0.18 ±0.07
eKalibr* [21]	492.34 ±14.99	492.32 ±14.93	321.11 ±5.94	237.72 ±5.23	−0.21 ±0.05
E2Calib [16]	504.28 ±10.44	504.36 ±10.50	320.72 ±5.90	234.97 ±4.36	−0.19 ±0.02
E2Calib‡	501.60 ±7.21	501.36 ±7.22	317.31 ±3.28	243.10 ±2.52	−0.18 **±0.01**
Ours	**498.86** **±4.17**	498.80 **±4.19**	**319.49** **±1.78**	**239.67** **±1.13**	**−0.20** ±0.02

*Circle-grid-based methods. ‡E2Calib using HyperE2VID [55].

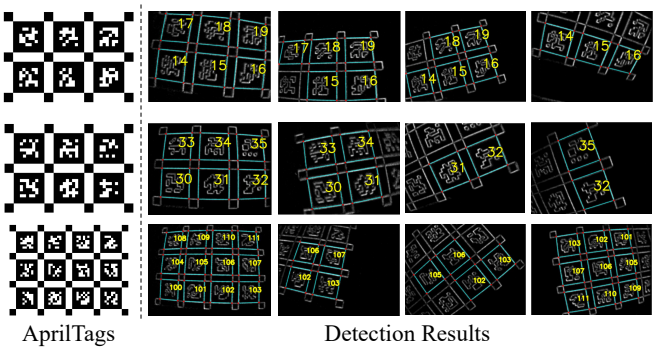


Fig. 13. Qualitative results on various AprilTags.

pose and camera trajectory for each corresponding sequence, changing only the texture applied to the calibration target. We generate 10 synthetic sequences for each target pattern.

We use the same reconstruction-based baselines as in the real-world experiments, namely E2Calib [16] with E2VID [40] and its variant using HyperE2VID [55]. In the synthetic setting, we additionally compare with the circle-grid-based EventCalib [20], E-Calib [22] and eKalibr [21].

4) Calibration Accuracy on Synthetic Data: We evaluate calibration accuracy by comparing the estimated intrinsic parameters with the known ground truth and report the mean and standard deviation across the 10 synthetic sequences. EventCalib [20] uses an inverse radial distortion model [57] whose coefficients are not directly comparable to the Brown–Conrady lens distortion coefficient k_1 [58]. Therefore, we do not report its distortion coefficient.

As shown in Table II, our method estimates the intrinsic parameters close to the ground truth while maintaining low variation across the sequences. Although E-Calib [22] provides a slightly closer mean estimate for f_y, it exhibits substantially larger variation. Overall, our method provides accurate intrinsic estimates with low variation across the synthetic sequences.

TABLE III
OUR STEREO CALIBRATION RESULTS.

Synthetic			**Real-world**	
e_R (°)	e_t (mm)	RMSE (px)	e_{rect} (px)	RMSE (px)
0.43 ± 0.28	2.94 ± 1.98	0.88	0.58 ± 0.19	0.78

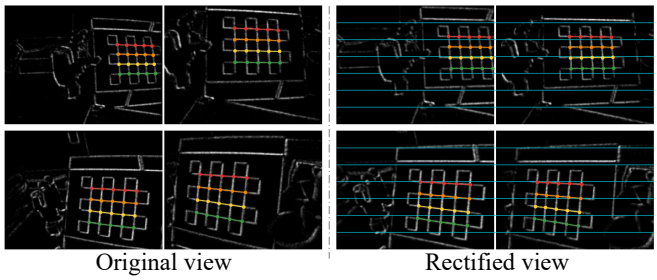


Fig. 14. Qualitative results on stereo calibration. The blue horizontal lines in the rectified views illustrate the alignment of corresponding checkerboard corners along the same image rows.

C. AprilTag Detection

Beyond checkerboards, we showcase that our approach extends to various fiducial markers such as AprilTag [15] (see Fig. 13). Our method successfully detects AprilTag with various IDs and geometric configurations, and handles partial board observations by correctly indexing the visible subset of tags even when others fall outside the field of view.

D. Stereo Calibration

We evaluate the proposed stereo calibration framework on both synthetic and real-world data. On synthetic data, where the ground-truth extrinsics are known, we measure the rotation error e_R [59] and translation error e_t [60] with respect to the ground truth. On real data, where ground-truth extrinsics are unavailable, we evaluate the geometric consistency of the estimated stereo extrinsics using the rectification error e_{rect}. For both synthetic and real data, we also report the stereo reprojection RMSE from OpenCV `stereoCalibrate(·)`.

For the synthetic experiment, we generate 10 sequences using a fixed stereo rig with known intrinsic and extrinsic parameters while varying the camera trajectory. We independently calibrate each sequence and report the resulting errors in Table III. The results show that our framework consistently estimates the relative pose of the stereo event camera across different camera trajectories.

For the real-world experiment, we use a rigidly mounted, hardware-synchronized stereo event camera and collect synchronized stereo event data. We first estimate the intrinsic parameters of the left and right cameras independently using all valid observations. We then randomly split the stereo observations into 70% for stereo calibration and 30% for evaluation. We use the remaining 30% to evaluate rectification consistency.

For the j-th checkerboard corner in the i-th evaluation observation, we compute the rectification error as

$$e_{\text{rect}}^{(i,j)} = \left| y_{L,j}^{\prime(i)} - y_{R,j}^{\prime(i)} \right|. \tag{48}$$

Here, y'_L and y'_R denote the vertical coordinates of the corresponding checkerboard corner in the rectified left and right images, respectively. We compute the mean and standard deviation of e_{rect} over all checkerboard correspondences in the evaluation observations. As shown in Table III, our method maintains an average vertical error of less than one pixel between corresponding rectified checkerboard corners, demonstrating consistent stereo geometry on observations not used for stereo extrinsic estimation. Fig. 14 shows qualitative stereo calibration results before and after rectification. After rectification, corresponding checkerboard corners are aligned to similar rows in the left and right views.

XI. Conclusion

We present the first event-based calibration framework that directly detects checkerboard corners without grayscale image reconstruction. Guided by a mathematical analysis revealing the absence of events at corner intersections, our method leverages edge-driven initialization and event density-based refinement to achieve sub-pixel corner accuracy. To handle parallel motion, we distinguish motion cases and recover suppressed edges using polarity information, enabling reliable corner initialization even when one edge becomes unobservable. Extensive experiments demonstrate that our approach achieves accurate calibration precision. By reintroducing checkerboards and extending them to AprilTag detection and stereo camera calibration, this work showcases the feasibility of real-world deployment for event camera calibration.

As a limitation, our method assumes a dominant global motion and estimates a single motion vector in IWE generation stage, which may degrade performance in scenes with spatially varying motion. Incorporating more flexible motion models (*e.g.,* pixel-wise motion) remains future work.

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